



## **Affirmed Networks Packaging and Deployment Solutions Guide**

Date: December 2022

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## Preface

This document describes how to deploy, upgrade, delete, operate, and maintain the Affirmed Networks Packaging and Deployment Solutions (PADS).

## Audience

This document is intended for network operators (sometimes referred to as Users or Operators in this document), installation staff, and other qualified service personnel who want to install, provision, monitor and learn about the Affirmed Networks system and associated features and functions. The audience should also be familiar with Long Term Evolution (LTE), and 3G or 4G networks. It may also be helpful to have a basic understanding of the HTTP protocol, TCP/IP stack, and the use of MMS services.

## Release Information

See the Release Notes for important information, requirements, limitations, and restrictions that apply to this release.

## How to Obtain Product Documentation

Documentation is available for currently supported product releases. You can access Affirmed Networks Guides on the [Manuals](#) site. From here, you can also download a PDF version of an individual guide, or the full manual set for a product. For access and instructions, please contact Support.

Other Affirmed Networks documentation may only be accessed from the SFTP server. This includes Release Notes, Lawful Intercept Documentation, and OpenSource Acknowledgments. Contact Support for instructions on how to access the SFTP server.

## Conventions

This guide uses the following conventions, when applicable:

| Description                                                                                | Convention                         | Examples                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Command names, keywords, dialog boxes, buttons, icons, menu                                | <b>Times New Roman bold font</b>   | Use <b>configure_ha</b> to configure...<br>Enter <b>y</b> .<br>To continue, press <b>Enter</b> .                                                                                                                              |
| Selecting a menu item                                                                      | <b>Menu =&gt; Option</b>           | Select <b>Security =&gt; Group Management</b>                                                                                                                                                                                 |
| Variable parameters for which you supply values                                            | <b>&lt;courier bold italic&gt;</b> | <b>configure_dr --host &lt;host name&gt;</b>                                                                                                                                                                                  |
| User input                                                                                 | <b>Courier bold font</b>           | <b>cd /opt/Affirmed/NMS/bin</b>                                                                                                                                                                                               |
| Screen messages, system output, sample source code                                         | Courier regular font               | result success                                                                                                                                                                                                                |
| Required keywords and arguments in square brackets                                         | [ ]                                | cluster [id] node [id] reboot                                                                                                                                                                                                 |
| Optional keywords and arguments in curly brackets                                          | { }                                | {cdr-file-name}                                                                                                                                                                                                               |
| Keywords separated by the pipe symbol. Requires you to specify only one item from the set. |                                    | Is this correct? [y n]                                                                                                                                                                                                        |
| Keywords representing a single element in angle brackets                                   | < >                                | <host name>                                                                                                                                                                                                                   |
| Asterisk after a CLI object indicates a required object                                    | *                                  | name*                                                                                                                                                                                                                         |
| Variables, book and chapter titles, file path names, new terms, and emphasized text        | <i>Times New Roman italic font</i> | See the <i>Affirmed Networks Packaging and Deployment Solutions Guide</i> .<br><i>Workflow</i> is an application running on the network element.                                                                              |
| Images in online help                                                                      |                                    | Many images can be enlarged by clicking the image with the left-mouse button. Alternately, they can be enlarged/reduced by pressing the Enter key when the image is highlighted (surrounded by a thick border) on the screen. |

**Note:** Additional information that may apply to the subject text.

**Tip:** Helpful suggestions.

**Caution:** Proceed carefully to avoid possible equipment damage or data loss.



## Customer Support

Affirmed Networks provides technical support 24 hours a day, 7 days a week (24 x 7) for any customer with an active maintenance contract.

[Email support](#)

[Open a support ticket](#)

### Call support

Local: +1-978-268-0800

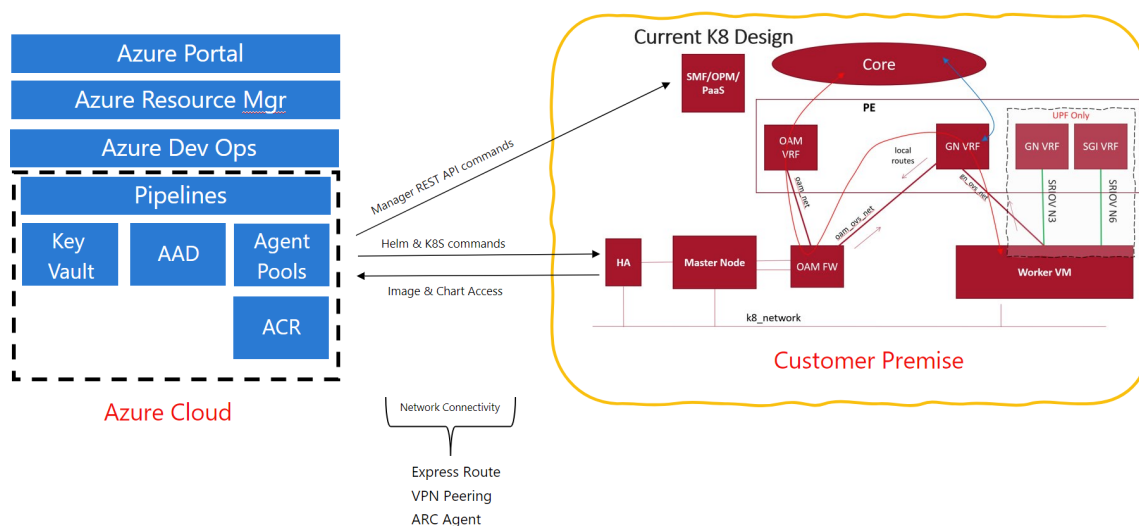
Toll-free: +1-888-283-2838



## Solution Design

Packaging and Deployment Solutions (PADS) is a method of using the Azure DevOps Service (ADO) as a deployment tool for UnityCloud. The ADO Pipeline is a continuous integration (CI) and continuous delivery (CD), testing, and deployment system. You can use the ADO release pipeline to automate the Kubernetes cluster finalization and deployment or upgrade the UnityCloud 5GC network-functions (NFs) – SMF (Session Management Function) and UPF (User Plane Function).

**Figure 1-1 PADS High Level Architecture**





## Configuring the Pipelines

PADS supports the following types of pipelines:

- Cluster building pipelines - used to finalize the cluster
- CNF deploy/delete pipelines - used for life-cycle management

ADO allows you to have a hosted agent and creates its own container on the agent using the specified image each time a pipeline is run. A self-hosted agent means that you provide a virtual machine (VM) or virtual machine scale set (VMSS) where the container is created at pipeline runtime. ADO enables the configured IP address and firewall rules to communicate to your on-premise resources.

### Agent Pool

This section describes how to create and configure an agent pool in your ADO Project, as well as how to install and run a self-hosted ADO agent.

**Figure 2-1 Agent Pools**



#### Run a self-hosted ADO agent

1. Create a pool in your ADO Project.
  - a. From the Project Settings tab, click **Add Pool** from the **Agent Pools** section.
  - b. Select **Self Hosted** and click **OK**.
2. Install a self-hosted ADO agent by creating a VM in your Resource Group to make the VM act as the agent pool:

```
mkdir -p ~/Downloads; cd ~/Downloads; wget
https://vstsagentpackage.azureedge.net/agent/2.191.1/vsts-
agent-linux-x64-2.191.1.tar.gz
```

```
mkdir -p ~/myagent && cd ~/myagent; tar zxvf ~/Downloads/vsts-agent-linux-x64-2.191.1.tar.gz
```

3. Register the self-hosted ADO agent by generating personal access tokens on ADO.

```
adoToken=<token>
```

```
adoOrganizationURL="https://dev.azure.com/A40-ADO-PADS-Dev"
```

```
adoAgentName=$<hostname>
```

*Note: You can regenerate the adoToken as needed.*

4. In the Agent Pools section, add the user who configures the permissions to set up an agent pool. The **adoAgentPoolName** is based on the pool name on ADO.

- a. From the Project Settings tab, click **Security** from the **Agent Pools** section.

- b. Select **Organization as administrator** and click **OK**.

```
adoAgentPoolName="pad-agentpool"
```

```
cd ~/myagent
```

```
./config.sh remove --unattended --token $<adoToken>
```

```
./config.sh --unattended --url $<adoOrganizationURL> --auth pat  
--token $<adoToken> --pool $<adoAgentPoolName> --agent  
$<adoAgentName> --acceptTeeEula
```

5. Run the self-hosted ADO agent.

```
cd ~/myagent
```

```
./run.sh
```

## Service Connection

A service connection is a connection between your ADO project and the resources, such as the Azure Container Registry (ACR), that are used by your pipeline. The service connection acts as a link between your ADO Project and your Azure Resource Group, so that ADO can recognize the location of your ACR and key vault.

You need two service connections for the PADS project: one to the Azure Resource Group and the other to the ACR—which is a Docker registry.

Figure 2-2 Edit Service Connection Window 1

**Edit service connection** ✕

Azure Resource Manager using service principal (manual)

**Environment**

Azure Cloud ▼

**Server Url**

https://management.azure.com/

**Scope Level**

☒ Subscription

☐ Management Group

☐ Machine Learning Workspace

**Subscription Id**

Subscription Id from the [publish settings file](#)

**Subscription Name**

a4opacketcore\_eng\_amc\_dev

Subscription Name from the [publish settings file](#)

**Authentication**

**Service Principal Id**

Client Id for connecting to the endpoint. Refer to [Azure Service Principal link](#) on how to create Azure Service Principal.

**Figure 2-3 Edit Service Connection Window (Continued)**

**Credential**

☒ Service principal key ☐ Certificate

**Service principal key**

\*\*\*\*\*

Service Principal Key for connecting to the endpoint. Refer to [Azure Service Principal link](#) on how to create Azure Service Principal. Ignore this field if the authentication type is spnCertificate.

**Tenant ID**

Tenant Id for connecting to the endpoint. Refer to [Azure Service Principal link](#) on how to create Azure Service Principal.

**Verify**

**Details**

**Service connection name**

pads-rg-sp

**Description (optional)**

**Security**

☒ Grant access permission to all pipelines

[Learn more](#) [Troubleshoot](#)

To connect your ADO project the Resource Group, you must provide an authentication ID and the necessary credentials using the Service Principal fields.



Figure 2-4 Edit Service Connection Window 2

**Edit service connection** [Close]

Registry type

☐ Docker Hub ☒ Others ☐ Azure Container Registry

Docker Registry

Docker ID

Docker Password

Email (optional)

Details

Service connection name

Description (optional)

Security

☒ Grant access permission to all pipelines

[Learn more](#) [Troubleshoot](#) Cancel Save

## Set Up Environments on ADO

You must add the following environments on ADO:

- CNF\_Deploy
- CNF\_Delete
- Finalize\_Cluster

- KubeConfig\_KeyVault

#### To create an environment in your ADO Project.

1. From the Pipelines tab, select Environments.
2. On the **Environments** section, click **New environment**.
3. Name the new environment.
4. (Optional) Provide a description and select a Resource.
5. Click **Create**.

## Configure the Pipelines

After you install the PADS software (see [Install PADS Software](#)), you can add and configure the variable values for the following pipelines:

- [ACR to OPM Image Copy Pipeline](#)
- [CNF Delete Pipeline](#)
- [CNF Deploy Pipeline](#)
- [Finalize K8s Cluster Pipeline](#)

### ACR to OPM Image Copy Pipeline

Configure the following files in the ACR to OPM Image Copy Pipeline and the corresponding repository (INFRA-DEV, LIB-CODE, or SVC-DEV).

#### ***acr-to-opm-image-copy-macro-pipeline.yml (INFRA-DEV)***

The following table shows the variables used in the ACR to OPM Image Copy macro pipeline.

**Table 2-1** acr-to-opm-image-copy-macro-pipeline.yml Variables

| Variables                                      | Description                                       |
|------------------------------------------------|---------------------------------------------------|
| <code>pool.name</code>                         | Name of the Agent Pool                            |
| <code>resources.repositories.repository</code> | Name of the repository used across the pipeline   |
| <code>resources.repositories.type</code>       | Type of repository                                |
| <code>resources.repositories.name</code>       | Name of the actual ADO repository                 |
| <code>resources.repositories.ref</code>        | Branch of the ADO repository                      |
| <code>resources.containers.container</code>    | Container reference name used across the pipeline |
| <code>resources.containers.image</code>        | Image name with reference to ACR                  |

| Variables                                                                                                                                                                                                                                                                                                                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>variables</b> <ul style="list-style-type: none"> <li>▪ <b>azureSubscription</b></li> <li>▪ <b>inputFile</b></li> <li>▪ <b>inputSourceFolder</b></li> <li>▪ <b>acrToOpmImageCopyTimeout</b></li> <li>▪ <b>spID</b></li> <li>▪ <b>spPassword</b></li> <li>▪ <b>sitespecific</b></li> <li>▪ <b>siteSpecificSourceFolder</b></li> </ul> | The following are ACR to OPM variables: <ul style="list-style-type: none"> <li>▪ Azure subscription name</li> <li>▪ Input JSON file name</li> <li>▪ Folder in LIB-CODE where inputFile is. Remove / at the end of the file path</li> <li>▪ Timeout in minutes for the pipeline</li> <li>▪ Azure Service Principal ID</li> <li>▪ Azure Service Principal Password. Should be protected</li> <li>▪ Site specific file name</li> <li>▪ Folder in INFRA-DEV where sitespecific is</li> </ul> |
| <b>stages.template</b>                                                                                                                                                                                                                                                                                                                 | Path to micro pipeline with the corresponding ADO repository                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>name</b>                                                                                                                                                                                                                                                                                                                            | Name of the Pipeline when it is run. Includes parameter zoneName                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>appendCommitMessageToRunName</b>                                                                                                                                                                                                                                                                                                    | Append the commit message to the build number. The default is true                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>zoneName</b>                                                                                                                                                                                                                                                                                                                        | Parameter to specify zone name                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>azureRegion</b>                                                                                                                                                                                                                                                                                                                     | Parameter for the Azure region the Pipeline runs on                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>versionSpecificSourceFolder</b>                                                                                                                                                                                                                                                                                                     | Directory path to version specific file                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>versionspecific</b>                                                                                                                                                                                                                                                                                                                 | Name of the version specific file                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### **code/acr\_opm\_image\_copy/acrToOpmCopy.json (LIB-CODE)**

The following table shows the variables used in the ACR to OPM Image Copy Input JSON pipeline.

**Table 2-2 acrToOpmCopy.json Variables**

| Variables                 | Description                                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>opm.ipAddress</b>      | OPM IP address                                                                                                 |
| <b>opm.sshUsername</b>    | SSH username used to communicate with OPM                                                                      |
| <b>opm.dockerRepo</b>     | Docker repository on OPM. For example:<br><b>&lt;ipaddress&gt;:&lt;port&gt;</b>                                |
| <b>opm.helmUrl</b>        | Destination helm URL for cna_deploy_inventory script. For now, just give OPM IP address                        |
| <b>opm.dockerYamlPath</b> | Docker YAML path for cna_deploy_inventory script. For now, just give <b>&lt;opm-ipaddress&gt;:&lt;port&gt;</b> |

| Variables                                     | Description                                                                                                |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <code>opm.pathToCopyPrds</code>               | File system path on OPM to where PRD files will be copied                                                  |
| <code>acrEndpoint</code>                      | ACR endpoint                                                                                               |
| <code>keyVault.name</code>                    | Azure Key Vault name                                                                                       |
| <code>keyVault.opmPrivateKeysecretName</code> | OPM secret name on the Azure Key-Vault for the private SSH key                                             |
| <code>helmChartsIgnore</code>                 | True if helm charts don't need to be imported. False otherwise                                             |
| <code>prdLocation</code>                      | Directory Path (location) to PRD files in LIB-CODE repository. Remove / at the end of the file path        |
| <code>prdFiles</code>                         | PRD files at "prdLocation" that must be processed                                                          |
| <code>additionalImages</code>                 | List of additional images that must be copied to OPM docker repository                                     |
| <code>cna_deploy_inventoryLocation</code>     | Directory Path where cna_deploy_inventory is stored. Remove / at the end of the file path                  |
| <code>cna_deploy_inventoryPathToCopy</code>   | File path on OPM to which cna_deploy_inventory script must be copied. Remove / at the end of the file path |

## CNF Delete Pipeline

This pipeline is used to delete the CNF (SMF/UPF) using the OPM. It deletes all the federations in the given K8s Cluster and can also perform a cleanup process after the deletion of all the federations on the K8s cluster. The cleanup process is an automation of the cleanup MOP, which is done manually to clean **rook-ceph**.

The following tables show the configurable files in the CNF Delete Pipeline and their corresponding (repository).

### ***cnf-operations-delete-macro-pipeline.yml (SVC-DEV)***

The following table shows the variables used in the CNF Delete macro pipeline.

**Table 2-3 CNF-operations-delete-macro-pipeline.yml Variables**

| Variables                                      | Description                                     |
|------------------------------------------------|-------------------------------------------------|
| <code>pool.name</code>                         | Name of the Agent Pool                          |
| <code>resources.repositories.repository</code> | Name of the repository used across the pipeline |
| <code>resources.repositories.type</code>       | Type of repository                              |

| Variables                                                                                                                                                                                                                                                                                                                                                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>resources.repositories.name</code>                                                                                                                                                                                                                                                                                                                       | Name of the actual ADO repository                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>resources.repositories.ref</code>                                                                                                                                                                                                                                                                                                                        | Branch of the ADO repository                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>resources.containers.container</code>                                                                                                                                                                                                                                                                                                                    | Container reference name used across the pipeline                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>resources.containers.image</code>                                                                                                                                                                                                                                                                                                                        | Image name with reference to ACR                                                                                                                                                                                                                                                                                                                                                                                                                         |
| variables <ul style="list-style-type: none"> <li>▪ <code>azureSubscription</code></li> <li>▪ <code>inputFile</code></li> <li>▪ <code>inputSourceFolder</code></li> <li>▪ <code>sitespecific</code></li> <li>▪ <code>siteSpecificSourceFolder</code></li> <li>▪ <code>cnfType</code></li> <li>▪ <code>spID</code></li> <li>▪ <code>spPassword</code></li> </ul> | The following are CNF variables: <ul style="list-style-type: none"> <li>▪ Azure subscription name</li> <li>▪ Input JSON file name</li> <li>▪ Folder in LIB-CODE where inputFile is. Remove / at the end of the file path.</li> <li>▪ Site specific file name</li> <li>▪ Folder in INFRA-DEV where sitespecific is.</li> <li>▪ SMF/UPF</li> <li>▪ Azure Service Principal ID</li> <li>▪ Azure Service Principal Password. Should be protected.</li> </ul> |
| <code>stages.template</code>                                                                                                                                                                                                                                                                                                                                   | Path to micro pipeline with the corresponding ADO repository                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>finalValidationDeleteTimeout</code>                                                                                                                                                                                                                                                                                                                      | Polling interval (in seconds) for checking pods and services status for each fed. Recommended value is 10                                                                                                                                                                                                                                                                                                                                                |
| <code>finalValidationTimeout</code>                                                                                                                                                                                                                                                                                                                            | Maximum timeout (in minutes) until which final polling happens for all pods and services. Following this, the pipeline exits with an error code if all pods and services are not running or complete. Recommended value is 120                                                                                                                                                                                                                           |
| <code>cnfDeleteCleanup</code>                                                                                                                                                                                                                                                                                                                                  | Cleanup process after CNF Delete. Value is <b>true</b> or <b>false</b>                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>cnfOperation</code>                                                                                                                                                                                                                                                                                                                                      | Operation of CNF. Keep it as <b>delete</b>                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>versionSpecificSourceFolder</code>                                                                                                                                                                                                                                                                                                                       | Directory path to version specific file                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>versionspecific</code>                                                                                                                                                                                                                                                                                                                                   | Name of the version specific file                                                                                                                                                                                                                                                                                                                                                                                                                        |

### ***cnf-operations-delete.json (LIB-CODE)***

The following table shows the variables used in the CNF Delete JSON pipeline.

**Table 2-4 CNF-operation-delete.json Variables**

| Variables           | Description                          |
|---------------------|--------------------------------------|
| <code>nfType</code> | Type of CNF being deployed (SMF/UPF) |

| Variables                                             | Description                                                                                                                                                                                               |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>kubeconfig.reference.keyVault.id</code>         | Name of the Azure Key-Vault                                                                                                                                                                               |
| <code>kubeconfig.reference.keyVault.secretName</code> | Name of the secret in Azure Key-Vault with value as kubeconfig of the K8s Cluster<br><b>&lt;cluster-name&gt;-kubeconfig</b>                                                                               |
| <code>opm.ipAddress</code>                            | OPM IP address                                                                                                                                                                                            |
| <code>opm.username</code>                             | Username used to authenticate to OPM application                                                                                                                                                          |
| <code>opm.keyVault.opmPasswordSecretName</code>       | Secret name in Azure Key-Vault of opm.username                                                                                                                                                            |
| <code>snapshotFolder</code>                           | File path in LIB-CODE to snapshot folder                                                                                                                                                                  |
| <code>snapshotFile</code>                             | Name of template snapshot file in LIB-CODE's <b>snapshotFolder</b>                                                                                                                                        |
| <code>snapshotVariableValuesFile</code>               | Name of snapshot variables/values file in snapshotFolder                                                                                                                                                  |
| <code>chartRepo</code>                                | Name of Chart Repository created on OPM. This is used by OPM to do Bulk Install using your snapshot                                                                                                       |
| <code>dzName</code>                                   | Name of Deployment Zone to which the K8s Cluster is attached                                                                                                                                              |
| <code>clusterId</code>                                | ID of the attached K8s Cluster from OPM. Retrieve this ID from the FinalizeK8sCluster Pipeline, Attach K8s Cluster stage. User can get this from the FinalizeK8sCluster Pipeline Attach K8s Cluster stage |
| <code>cephDiskType</code>                             | Takes values either <b>vdc</b> for SMF or <b>vdb</b> for UPF                                                                                                                                              |

## CNF Deploy Pipeline

The following tables show the configurable files in the CNF Deploy Pipeline and their corresponding (repository).

### **cnf-operations-deploy-macro-pipeline.yml (SVC-DEV)**

The following table shows the variables used in the CNF Deploy macro pipeline.

**Table 2-5 CNF-operations-deploy-macro-pipeline.yml Variables**

| Variables                                      | Description                                     |
|------------------------------------------------|-------------------------------------------------|
| <code>pool.name</code>                         | Name of the Agent Pool                          |
| <code>resources.repositories.repository</code> | Name of the repository used across the pipeline |

| Variables                                                                                                                                                                                                                                                                                                                                                                                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>resources.repositories.type</code>                                                                                                                                                                                                                                                                                                                                                                | Type of repository                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>resources.repositories.name</code>                                                                                                                                                                                                                                                                                                                                                                | Name of the actual ADO repository                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>resources.repositories.ref</code>                                                                                                                                                                                                                                                                                                                                                                 | Branch of the ADO repository                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>resources.containers.container</code>                                                                                                                                                                                                                                                                                                                                                             | Container reference name used across the pipeline                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>resources.containers.image</code>                                                                                                                                                                                                                                                                                                                                                                 | Image name with reference to ACR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| variables <ul style="list-style-type: none"> <li>▪ <code>azureSubscription</code></li> <li>▪ <code>inputFile</code></li> <li>▪ <code>inputSourceFolder</code></li> <li>▪ <code>sitespecific</code></li> <li>▪ <code>siteSpecificSourceFolder</code></li> <li>▪ <code>cnfType</code></li> <li>▪ <code>cnfDeployTimeout</code></li> <li>▪ <code>spID</code></li> <li>▪ <code>spPassword</code></li> </ul> | The following are CNF variables: <ul style="list-style-type: none"> <li>▪ Azure subscription name</li> <li>▪ Input JSON file name</li> <li>▪ Folder in LIB-CODE where inputFile is. Remove / at the end of the file path.</li> <li>▪ Site specific file name</li> <li>▪ Folder in INFRA-DEV where sitespecific is.</li> <li>▪ smf/upf</li> <li>▪ Total timeout in minutes for the CNF pipeline to run.</li> <li>▪ Azure Service Principal ID</li> <li>▪ Azure Service Principal Password. Should be protected.</li> </ul> |
| <code>stages.template</code>                                                                                                                                                                                                                                                                                                                                                                            | Path to micro pipeline with the corresponding ADO repository                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>name</code>                                                                                                                                                                                                                                                                                                                                                                                       | Name of the Pipeline when it is run. Includes parameter <b>zoneName</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>appendCommitMessageToRunName</code>                                                                                                                                                                                                                                                                                                                                                               | Append the commit message to the build number. The default is true                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>fedDeployTimeout</code>                                                                                                                                                                                                                                                                                                                                                                           | Polling interval (in seconds) for calling NB API to poll all federations status. Recommended value is 10                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>zoneName</code>                                                                                                                                                                                                                                                                                                                                                                                   | Parameter to specify zone name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>workerCount</code>                                                                                                                                                                                                                                                                                                                                                                                | Parameter for number of workers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>azureRegion</code>                                                                                                                                                                                                                                                                                                                                                                                | Parameter for the Azure region on which the Pipeline runs                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>versionSpecificSourceFolder</code>                                                                                                                                                                                                                                                                                                                                                                | Directory path to version specific file                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>versionspecific</code>                                                                                                                                                                                                                                                                                                                                                                            | Name of the version specific file                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

### ***code/cnf\_operations/cnf-operations-deploy.json (LIB-CODE)***

The following table shows the variables used in the CNF Deploy Input JSON pipeline.

Table 2-6 CNF-operation-deploy.json Variables

| Variables                                             | Description                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>nfType</code>                                   | Type of CNF being deployed (SMF/UPF)                                                                                                                                                                                                                                                          |
| <code>kubeconfig.reference.keyVault.id</code>         | Name of the Azure Key-Vault                                                                                                                                                                                                                                                                   |
| <code>kubeconfig.reference.keyVault.secretName</code> | Name of the secret in Azure Key-Vault with value as kubeconfig of the K8s Cluster<br><b>&lt;cluster-name&gt;-kubeconfig</b>                                                                                                                                                                   |
| <code>extensionPrdFolder</code>                       | Directory Path (location) to extension PRD files in the LIB-CODE repository. Remove / at the end of the file path<br><br><i>Note: Because the start/stop order of the extension PDR file changes based on the UC version, keep this folder inside the appropriate UC version PRDs folder.</i> |
| <code>extensionPrdFile</code>                         | Extension PRD file to use the start/stop order for CNF deployment                                                                                                                                                                                                                             |
| <code>opm.ipAddress</code>                            | OPM IP address                                                                                                                                                                                                                                                                                |
| <code>opm.username</code>                             | Username used to authenticate to OPM application                                                                                                                                                                                                                                              |
| <code>opm.keyVault.opmPasswordSecretName</code>       | Secret name in Azure Key-Vault of opm.username                                                                                                                                                                                                                                                |
| <code>snapshotFolder</code>                           | File path in LIB-CODE to snapshot folder                                                                                                                                                                                                                                                      |
| <code>snapshotFile</code>                             | Name of template snapshot file in LIB-CODE's <b>snapshotFolder</b>                                                                                                                                                                                                                            |
| <code>snapshotVariableValuesFile</code>               | Name of snapshot variables/values file in snapshotFolder                                                                                                                                                                                                                                      |
| <code>chartRepo</code>                                | Name of Chart Repository created on OPM. This is used by OPM to do Bulk Install using your snapshot                                                                                                                                                                                           |
| <code>dzName</code>                                   | Name of Deployment Zone to which the K8s Cluster is attached                                                                                                                                                                                                                                  |
| <code>clusterId</code>                                | ID of the attached K8s Cluster from OPM. Retrieve this ID from the FinalizeK8sCluster Pipeline, Attach K8s Cluster stage                                                                                                                                                                      |

**code/cnf\_operations/snapshots/snapshot\_specific\_values.json (LIB-CODE)**

This file contains the values to the variables in the snapshot file used by the CNF Deploy Pipeline.

Use the following example to assign values to variables, such as `$ml_peer` and `$peer_as`:

```
{
```



```

"$ml_peer": "10.161.30.8",
"$peer_as": "64500"
}

```

## Finalize K8s Cluster Pipeline

The following tables show the configurable files in the Finalize K8s Cluster Pipeline and their corresponding (repository).

### *finalizeK8sCluster-macro-pipeline.yml (INFRA-DEV)*

The following table shows the variables used in the FinalizeK8sCluster macro pipeline.

Table 2-7 finalizeK8sCluster-macro-pipeline.yml Variables

| Variables                                                                                                                                                                                                                                                                                                                                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>pool.name</code>                                                                                                                                                                                                                                                                                                                                          | Name of the Agent Pool                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>resources.repositories.repository</code>                                                                                                                                                                                                                                                                                                                  | Name of the repository used across the pipeline                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>resources.repositories.type</code>                                                                                                                                                                                                                                                                                                                        | Type of repository                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>resources.repositories.name</code>                                                                                                                                                                                                                                                                                                                        | Name of the actual ADO repository                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>resources.repositories.ref</code>                                                                                                                                                                                                                                                                                                                         | Branch of the ADO repository                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>resources.containers.container</code>                                                                                                                                                                                                                                                                                                                     | Container reference name which will be used across the pipeline                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>resources.containers.image</code>                                                                                                                                                                                                                                                                                                                         | Image name with reference to ACR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| variables <ul style="list-style-type: none"> <li>▪ <code>azureSubscription</code></li> <li>▪ <code>inputFile</code></li> <li>▪ <code>inputSourceFolder</code></li> <li>▪ <code>sitespecific</code></li> <li>▪ <code>siteSpecificSourceFolder</code></li> <li>▪ <code>spID</code></li> <li>▪ <code>spPassword</code></li> <li>▪ <code>spTenant</code></li> </ul> | The following are Kubernetes Cluster variables: <ul style="list-style-type: none"> <li>▪ Azure subscription name</li> <li>▪ Input JSON file name</li> <li>▪ Folder in LIB-CODE where inputFile is. Remove / at the end of the file path</li> <li>▪ Site specific file name</li> <li>▪ Folder in INFRA-DEV containing the <code>sitespecific</code> variable</li> <li>▪ Azure Service Principal ID</li> <li>▪ Protected Azure Service Principal Password</li> <li>▪ Protected Azure Service Principal Tenant ID</li> </ul> |
| <code>stages.template</code>                                                                                                                                                                                                                                                                                                                                    | Path to micro pipeline with the corresponding ADO repository                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>name</code>                                                                                                                                                                                                                                                                                                                                               | Name of the Pipeline when it is run. Includes parameter <code>zoneName</code>                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Variables                                 | Description                                                                                     |
|-------------------------------------------|-------------------------------------------------------------------------------------------------|
| <code>appendCommitMessageToRunName</code> | Append the commit message to the build number. The default is true                              |
| <code>zoneName</code>                     | Parameter to specify zone name                                                                  |
| <code>workerCount</code>                  | Parameter for number of workers                                                                 |
| <code>azureRegion</code>                  | Parameter for the Azure region on which the Pipeline runs                                       |
| <code>forceLocalDNS</code>                | Parameter to fix local DNS issue with VM images                                                 |
| <code>versionSpecificSourceFolder</code>  | Directory path to version specific file                                                         |
| <code>versionspecific</code>              | Name of the version specific file                                                               |
| <code>forcelocalDnsPods</code>            | True/False parameter if fix for force local DNS pods must be applied after K8s Cluster creation |

### ***code/finalize\_cluster/finalizeK8sCluster.json (LIB-CODE)***

The following table shows the variables used in the FinalizeK8sCluster Input JSON pipeline.

**Table 2-8 finalizeK8sCluster.json Variables**

| Variables                                                                        | Description                                                                                                               |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <code>clusterName</code>                                                         | Name of the K8s cluster used for kubeconfig                                                                               |
| <code>clusterParameters.ingressVM.adminAccess</code>                             | Communication type to Cluster. Accept the default value, SSH                                                              |
| <code>ClusterParameters.ingressVM.publicIpAddress</code>                         | IP-address to communicate with the jump VM                                                                                |
| <code>ClusterParameters.ingressVM.port</code>                                    | SSH port number to communicate with the jump VM                                                                           |
| <code>ClusterParameters.ingressVM.adminUsername</code>                           | SSH username to communicate with the jump VM. Accept the default value, root                                              |
| <code>ClusterParameters.ingressVM.adminPrivateKey.reference.keyVault.name</code> | Azure Key-Vault name with SSH private-key to be used to get to the ingress-VM                                             |
| <code>clusterParameters.ingressVM.AdminPrivateKey.reference.secretName</code>    | Azure Key-Vault secret-name where the SSH private-key is used to get to the location of the VM                            |
| <code>clusterParameters.tuning</code>                                            | True if Cluster tuning needs to be done. False otherwise                                                                  |
| <code>ClusterParameters.vm2MKey.reference.keyVault.name</code>                   | Azure Key-Vault name with SSH private-key to be used from the ingress-VM to the individual VMS that forms the K8s cluster |

| Variables                                                   | Description                                                                                                                                    |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>ClusterParameters.vm2MKey.reference.secretName</code> | Azure Key-Vault secret-name where the SSH private-key to be used from the ingress-VM to the individual VMs that form the K8s cluster is stored |
| <code>clusterParameters.vms[].ipAddress</code>              | VM IP address                                                                                                                                  |
| <code>clusterParameters.vms[].ipAddressInt</code>           | VM Internal IP address                                                                                                                         |
| <code>clusterParameters.vms[].port</code>                   | VM SSH Port                                                                                                                                    |
| <code>clusterParameters.vms[].type</code>                   | Type of Node (Master/Worker)                                                                                                                   |
| <code>clusterParameters.vms[].adminUsername</code>          | SSH username to communicate with VMs                                                                                                           |
| <code>clusterParameters.vms[].dataplane</code>              | Set to <b>true</b> if the node (VM) must have a label <b>nodetype=dataplane</b> , otherwise set this value to <b>false</b>                     |
| <code>opm.ipAddress</code>                                  | OPM IP address                                                                                                                                 |
| <code>opm.sshUsername</code>                                | SSH username to communicate to OPM                                                                                                             |
| <code>opm.keyVault.name</code>                              | Azure Key-Vault name with OPM information                                                                                                      |
| <code>opm.keyVault.opmPrivateKeysecretName</code>           | OPM secret name on the Azure Key-Vault for the private SSH key                                                                                 |
| <code>opm.certPath</code>                                   | Certificate path on OPM which copies to all VMs on K8s Cluster                                                                                 |
| <code>certCopyPathToAllVMs</code>                           | Path on each VM in cluster to where the certificate copies                                                                                     |
| <code>opm.username</code>                                   | OPM GUI username                                                                                                                               |
| <code>opm.opmPasswordSecretName</code>                      | Azure Key-Vault secret name for OPM GUI user's password                                                                                        |
| <code>opm.dzName</code>                                     | Deployment Zone name on OPM to which the create K8s Cluster needs to be attached                                                               |

# Running Pipelines

You must meet the following prerequisites in order to properly run the configured pipelines, including FinalizeK8sCluster, CNF Delete, and CNF Deploy Pipelines.

## Prerequisites to Initially Run Pipelines

PADS 1.4.3 uses the Operations and Policy Manager (OPM) for CNF Operations such as deploy, delete, update.

Since we use NB API to communicate to the OPM, you must update the OPM's default SSL Certificate to include the OPM's IP Address. Complete the following steps to update the SSL Certificate:

***Note:** The following prerequisites to initially run pipelines only need to be completed once. Once these are completed, they are applied to all pipeline runs subsequently created.*

1. SSH into the OPM VM.
2. Create a new certificate for the OPM, which contains the OPM IP address.  

```
openssl req -x509 -newkey rsa:4096 -keyout key.pem -out cert.pem -subj '/CN=ambassador-cert' -nodes -addext "subjectAltName = IP:<opm-ip-address>"
```
3. Print the certificate and verify an IP address is provided.  

```
openssl x509 -in cert.pem -text
```
4. Create a Kubernetes secret with cert.pem as its value.  

```
kubectl create secret tls tls-cert --cert=cert.pem --key=key.pem -n vnfmgr
```
5. Verify the secret is created.  

```
kubectl get secret -n vnfmgr
```
6. Edit the wildcard-host of the OPM and change the `spec.tlsSecret.name` value to `tls-cert`.  

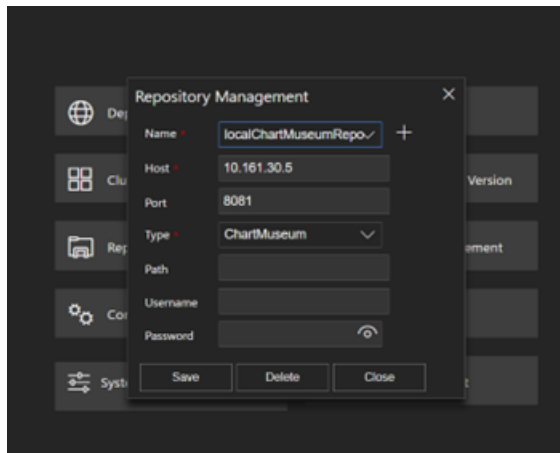
```
kubectl edit host wildcard-host -n vnfmgr
```
7. Log in to the OPM GUI and verify the new certificate works correctly.

## Prerequisites to Run a FinalizeK8sCluster Pipeline

Complete the following prerequisites prior to running a FinalizeK8sCluster Pipeline.

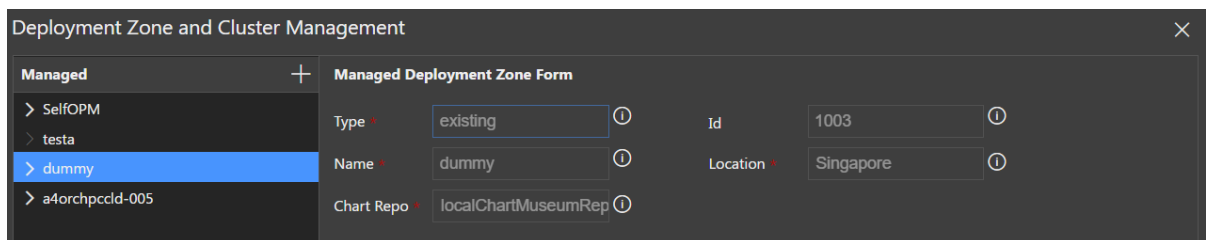
1. Create Chart Repository on OPM (Repository Management) using the details of OPM Chart Museum.

Figure 2-5 OPM - Create Chart Repository



2. Create an existing Deployment Zone by providing the Deployment Zone name in the finalsizeK8sClutser.json file and the Chart Repository you created in the previous step.

Figure 2-6 OPM - Create Deployment Zone



## Prerequisites to Run a CNF Deploy Pipeline

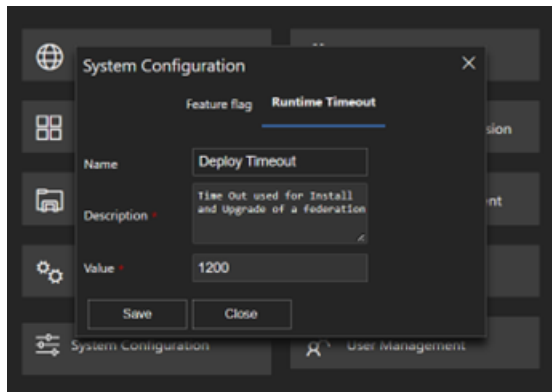
Complete the following prerequisite steps prior to running a CNF Deploy Pipeline.

1. Load all of the Helm Charts to OPM's Chart Museum Repository using the cna\_deploy\_inventory script.

```
./cna_deploy_inventory --inventory *.json --dest_docker_url
https://$<OPM>:5005 --dest_helm_url https://$<OPM>/chartm/ --
docker_yaml_path ${OPM}:5005 --login <user>:<password>
```

**Note:** You must run the CNF deploy Pipeline on a K8s Cluster with 0 federations installed.

2. Update the **Runtime Deploy Timeout** value in the OPM GUI. The recommended value is **1200**.

**Figure 2-7 OPM - Update Runtime Deploy Timeout**

3. Get the Cluster ID from the FinalizeK8sCluster Pipeline, attach the K8s Cluster Stage log and update it in *cnf-operations-deploy.json*.

**Figure 2-8 OPM - FinalizeK8sCluster ClusterID**

|                                   |     |
|-----------------------------------|-----|
| ✓ Access OPM User Password fr...  | <1s |
| ✓ Set variable to OPM SSH Pass... | <1s |
| ✓ Downloading OPM Public Cer...   | <1s |
| ✓ Attach K8 Cluster to OPM        | 4s  |
| ✓ Delete SSH Private Key          | <1s |
| ✓ Delete Source Folders           | 10s |
| ✓ Post-job: Checkout LIB-CODE...  | <1s |
| ✓ Post-job: Checkout INFRA-DE...  | <1s |
| ✓ Stop Containers                 | <1s |

```

19 22:49PM UTC on Oct 31, 2022
20 Authorization Token not set.
21 Getting Authorization Header token from OPM: 10.161.30.5
22 Successfully retrieved Authorization Token for OPM
23
24 -----
25 Attaching K8 cluster to OPM.
26 Attaching cluster to OPM 10.161.30.5 and Deployment Zone 'dummy' using file: /tmp/690/attachK8Cluster2/kubeconfig
27 Cluster (5f7342.a7586ab7c67467ed) attached successfully using config file to stack dummy
28 Successfully attached Cluster to OPM.
29
30 -----
31 Attached K8 Cluster ID: 5f7342.a7586ab7c67467ed
32
33 -----
34 Finishing: Attach K8 Cluster to OPM

```

4. Get the Templatized Snapshot with variables, copy it to LIB-CODE and update the snapshot file path in *cnf-operations-deploy.json*.
5. Add the list of variables as keys and corresponding values in the *snapshot\_specific\_values.json* file.
6. Create a secret in Azure Key-Vault with an **<OPM NB API user and password>** value.

## Run a Kubernetes Cluster Building Pipeline

Prior to running a Kubernetes Cluster Building pipeline, verify that the prerequisite tasks are complete. See [Prerequisites to Run a FinalizeK8sCluster Pipeline](#).

### Running a Kubernetes Cluster Building Pipeline

1. Access the **Pipelines** page.
2. On the list of pipelines, select the **[your org] finalize-K8s** pipeline.
3. Click the **Run pipeline** button.
4. On the **Run pipeline** window:

- a. Verify the correct branch is displayed or select it from the drop-down.
- b. Verify the advanced options are configured correctly.
- c. (Optional) Select to enable system diagnostics.
- d. Click **Run**.

## Run a CNF Deploy Pipeline

Prior to running a CNF Deploy pipeline, verify that the prerequisite tasks are complete. See [Prerequisites to Run a CNF Deploy Pipeline](#).

### Running a CNF Deploy Pipeline

1. Access the **Pipelines** page.
2. On the list of available pipelines, select the **[your org] CNFDeploy** pipeline.
3. Click the **Run pipeline** button.
4. On the **Run pipeline** window:
  - a. Verify the correct branch is chosen or select it from the drop-down.
  - b. Verify the advanced options are configured correctly.
  - c. (Optional) Select to enable system diagnostics.
  - d. Click **Run**.

## Run a CNF Delete Pipeline

Prior to running a CNF Delete pipeline, verify that the prerequisite tasks are complete.

### Running a CNF Delete Pipeline

1. Access the **Pipelines** page.
2. On the list of available pipelines, select the **[your org] CNFDelete** pipeline.
3. Click the **Run pipeline** button.
4. On the **Run pipeline** window:
  - a. Verify the correct branch is chosen or select it from the drop-down.
  - b. Verify the advanced options are configured correctly.
  - c. (Optional) Select to enable system diagnostics.
  - d. Click **Run**





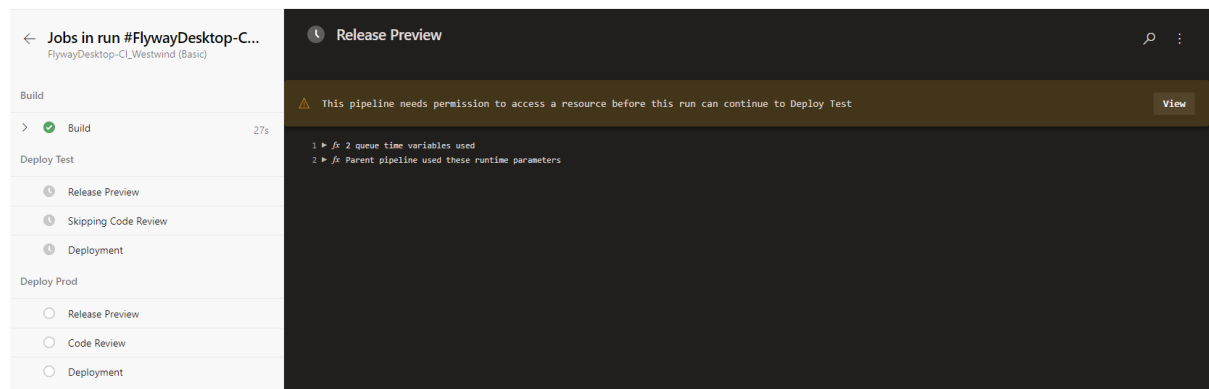
## Operations and Maintenance

Some of the ADO packages required to run the ADO tasks in the pipelines are not available by default and you must obtain them from the Visual Studio marketplace. Use the following links to install the specific packages:

- [JSON to Variable - task that reads JSON file in your build agent to generate Build/Release variables to be used in other steps](#)
- [Replace Tokens - extension that replaces tokens in text files with variable values](#)
- [Variable Toolbox - Azure Pipelines tasks for setting and expanding variables](#)

When running the Pipelines for the first time, you must give explicit permission to access other repositories like LIB-CODE.

**Figure 3-1 Release Preview**



**Note:** This is only needed for the initial run. All subsequent runs of the same pipeline will re-use the granted permissions from the first run.

If any pods or services fail to come up, use the following commands to debug and get logs:

- List all pods and their status  
— **kubectl get pods -A**

- Describe the status of a specific pod in more detail. Usually used to look into a failed pod.  
— **kubectl describe pod <pod> -n <namespace>**
- List all services and their status  
— **kubectl get svc -A**
- Describe the status of a specific service in more detail. Usually used to look into a failed service.  
— **kubectl describe svc <service> -n <namespace>**

When there is an error in a pipeline with any of the operations (OPM-related) in the following table and there is no error message related to the error, use the following command to examine the logs of the OPM on the OPM VM:

**kubectl logs <api-gw-pod> -n vnfmgr**

The following table shows the OPM-related operations in which an error may occur without displaying a message related to the error, along with the strings you can search for to identify the reasons for the errors.

**Table 3-1 NB API Strings to Debug**

| NB API                                                                          | String to help debug    |
|---------------------------------------------------------------------------------|-------------------------|
| Attaching pre-existing Cluster (Finalize K8s Cluster Pipeline)                  | attachExistingCluster   |
| Import NF Config Set and Snapshot (CNF Deploy Pipeline)                         | importNFConfigSet       |
| Bulk Installation (CNF Deploy Pipeline)                                         | installNetworkFunctions |
| <a href="#">Get All Federation Status</a> (CNF Deploy Pipeline)                 | getAllFedStatus         |
| <a href="#">Get Single Federation pods running status</a> (CNF Deploy Pipeline) | retrieveFedPodsStatus   |

## Install PADS Software

### Prerequisites

Verify that the following prerequisite tasks are complete prior to installing the PADS software:

- Extract the `pads_pipeline-<version>-<build_number>.tgz` file. Extract the `pipeline_artifacts.tar.gz` file inside `pads_pipeline-<version>-<build_number>/`.
  - The results should show the 3 directories LIB-CODE/, INFRA-DEV/ and SVC-DEV/ extracted.
- Confirm that the following repositories exist on the ADO project: LIB-CODE, INFRA-DEV, and SVC-DEV.
- Confirm that Service Connections to the following exist on the ADO project: Azure Resource Group and Azure Container Registry.

### Installing the PADS Software

Complete the following steps in each ADO repository.

#### LIB-CODE

1. Copy the contents of the *LIB-CODE/* directory extracted from *pipeline\_artifacts.tar.gz* into the LIB-CODE ADO repository.
2. Update the *code/acr\_opm\_image\_copy/acrToOpmCopy.json* file with appropriate values.
3. Update the *code/finalize\_cluster/finalizeK8sCluster.json* file with appropriate values.
4. Update the following files with the appropriate files.
  - *code/cnf\_operations/cnf-operations-deploy.json*
  - *code/cnf\_operations/cnf-operations-delete.json*
5. Add the *cna\_deploy\_inventory* script file to a folder in LIB-CODE and refer to it in the appropriate variables.

6. Add the *snapshotFile* value and *snapshot\_specific\_values.json* files to LIB-CODE and refer to them in the appropriate variables.

### **INFRA-DEV**

1. Copy the contents of the INFRA-DEV/ directory extracted from *pipeline\_artifacts.tar.gz* into the INFRA-DEV ADO repository.
2. Update the values in the following macro pipeline files: *acr-to-opm-image-copy-macro-pipeline.yml* and *finalizeK8sCluster-macro-pipeline.yml*.

### **SVC-DEV**

1. Copy the contents of the SVC-DEV/ directory extracted from *pipeline\_artifacts.tar.gz* into the SVC-DEV ADO repository.
2. Update the values in the macro pipeline files, *cnf-operations-delete-macro-pipeline.yml* and *cnf-operations-deploy-macro-pipeline.yml*.

## Upgrade PADS Software

### Prerequisites

Verify that the following prerequisite tasks are complete prior to upgrading the PADS software:

- Extract the `pads_pipeline-<version>-<build_number>.tgz` file. Extract the `pipeline_artifacts.tar.gz` file inside `pads_pipeline-<version>-<build_number>/`.
  - The results should show the 3 directories LIB-CODE/, INFRA-DEV/ and SVC-DEV/ extracted.
- Confirm that the following repositories exist on the ADO project: LIB-CODE, INFRA-DEV, and SVC-DEV.
- Confirm that Service Connections to the following exist on the ADO project: Azure Resource Group and Azure Container Registry.

### Steps to Upgrade

Complete the following steps in each ADO repository to upgrade.

#### LIB-CODE

1. Override all files in `pipeline/` with the contents of `LIB-CODE/pipeline/` directory extracted from `pipeline_artifacts.tar.gz`.
2. Add new variables to `code/acr_opm_image_copy/acrToOpmCopy.json`.
  - Use `LIB-CODE/code/acr_opm_image_copy/acrToOpmCopy.json` extracted from `pipeline_artifacts.tar.gz` for reference.
3. Add new variables to `code/finalize_cluster/finalizeK8sCluster.json` file.
  - Use `LIB-CODE/code/finalize_cluster/finalizeK8sCluster.json` extracted from `pipeline_artifacts.tar.gz` for reference.
4. Update below files with appropriate variables from Release Notes. Use the corresponding LIB-CODE/ directory files for reference.

- `code/cnf_operations/cnf-operations-deploy.json`
  - `code/cnf_operations/cnf-operations-delete.json`
5. Add the *cna\_deploy\_inventory* script file to a folder in LIB-CODE and refer to the script file.
  6. Add the PRD files to *code/acr\_opm\_image\_copy* and refer to the PRD files.

## INFRA-DEV

Update the values in the following macro pipeline files: *acr-to-opm-image-copy-macro-pipeline.yml* and *finalizeK8sCluster-macro-pipeline.yml*.

## SVC-DEV

Update the values in the macro pipeline files, *cnf-operations-delete-macro-pipeline.yml* and *cnf-operations-deploy-macro-pipeline.yml*.